

Jyothir Raghavalu Bhogi

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SUMMARY

Applied AI Engineer focused on building and evaluating useful AI systems with Python, PyTorch, LLMs, Generative AI, and modern ML frameworks. Hands-on experience developing RAG and agentic systems, designing retrieval and evaluation pipelines, benchmarking model behavior, and experimenting with AI workflows. Strong foundation in machine learning, deep learning, NLP, model evaluation, and backend engineering, with a focus on rigorous experimentation, performance, and building reliable AI systems.

EDUCATION

Lovely Professional University

B.Tech in Computer Science and Engineering

Jalandhar, Punjab

Aug 2022 – May 2026

- **Relevant Coursework:** Artificial Intelligence, Machine Learning, Deep Learning, NLP, Data Structures, Algorithms, DBMS, Distributed Systems

EXPERIENCE

AI Intern

Know AI

June 2026 – Present

Ahmedabad, Gujarat

- Build and evaluate production-oriented AI systems using Python, PyTorch, Generative AI, LLMs, and modern machine learning frameworks.
- Develop LLM and RAG workflows involving embeddings, retrieval, context construction, prompt experimentation, and model response evaluation.
- Design AI experiments to analyze model behavior, retrieval quality, response reliability, and performance across different configurations.
- Implement agentic workflows and tool-integrated AI systems with emphasis on model utility, reasoning workflows, and reliable task execution.
- Optimize AI pipelines through systematic debugging, benchmarking, experimentation, and performance analysis.

Data Engineer Intern

flickDone

June 2025 – Aug 2025

Remote (South Korea-based company)

- Built scalable Python and SQL data pipelines processing 50,000+ records for analytics and machine learning workflows.
- Developed data processing pipelines using Python, Pandas, and NumPy for ingestion, transformation, validation, and preparation of structured datasets.
- Implemented automated data validation workflows improving the reliability and consistency of downstream analytical and ML workloads.
- Optimized data processing and query workflows, improving pipeline reliability and reducing data access latency.
- Contributed to automated engineering workflows using Git, Docker, CI/CD, and Linux development environments.

PROJECTS

Regulatory Compliance RAG Assistant | *Python, ChromaDB, BM25, Hugging Face*

Apr 2026 – Jun 2026

- Built and evaluated a retrieval-augmented AI system for regulatory document understanding using Python, Hugging Face, ChromaDB, and BM25.
- Designed hybrid retrieval combining dense vector search and BM25, using Reciprocal Rank Fusion to improve document retrieval quality.
- Implemented cross-encoder reranking and LLM-based generation to improve context relevance and produce grounded responses with citations.
- Benchmarked retrieval strategies using Hit Rate and Mean Reciprocal Rank across BM25, vector, hybrid, and reranked configurations.

AI Product Recommendation System | *Python, FastAPI, FAISS, LLMs*

Jun 2025 – Aug 2025

- Built an embedding-based recommendation system using Python, FAISS, BGE embeddings, and LLMs for semantic product retrieval.

- Implemented dense vector search over product metadata and evaluated retrieved candidates for relevance to natural-language queries.
- Integrated Claude 3 Haiku through OpenRouter to rerank retrieved candidates and improve recommendation quality.
- Developed an evaluation-oriented workflow for comparing retrieval and reranking behavior across product queries.

Multi-Stage Ensemble for Driver Fatigue Recognition | *Python, PyTorch, Scikit-learn, Deep Learning* 2025

- Developed a multi-stage ensemble deep learning system for driver fatigue recognition, achieving 96.8% classification accuracy.
- Designed and evaluated multiple model stages for extracting complementary behavioral and visual features for fatigue classification.
- Experimented with model architectures, feature representations, and ensemble strategies to improve predictive performance.
- Analyzed model performance using classification metrics and iterative experimentation to identify effective configurations.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++

Machine Learning: Machine Learning, Deep Learning, NLP, Model Evaluation, Benchmarking, Retrieval Systems, Representation Learning

Generative AI: LLMs, Generative AI, RAG, Agentic AI, Prompt Engineering, Context Management, Embeddings, Reranking

Frameworks: PyTorch, Hugging Face, LangChain, LangGraph, FastAPI, Scikit-learn

AI Systems: Model Evaluation, AI Experimentation, Retrieval Pipelines, Agent Orchestration, AI Workflows, Performance Optimization

Data & Infrastructure: FAISS, ChromaDB, Vector Databases, PostgreSQL, MySQL, Pandas, NumPy, Docker, Linux

Engineering: REST APIs, Git, GitHub Actions, CI/CD, Testing, Debugging, Workflow Automation